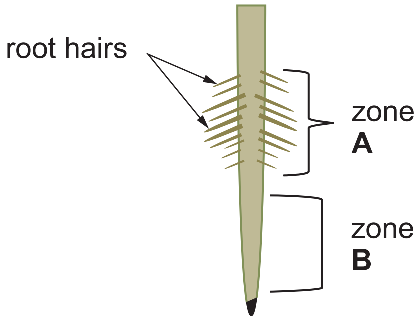
Examiner only

Answer **all** questions.

1. Part of a plant root is shown in **Image 1.1**.

Image 1.1



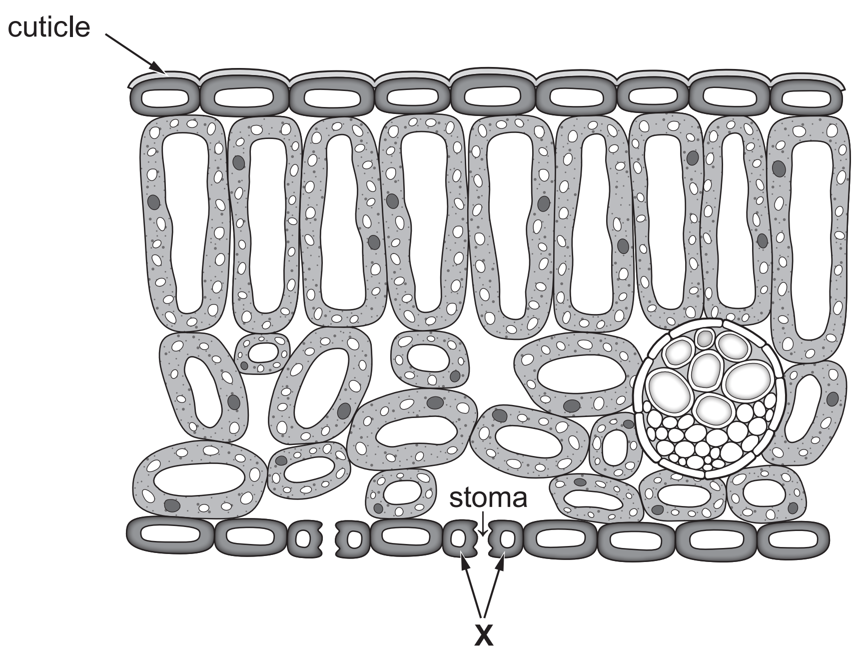
(a) Explain why water uptake in zone **A** is greater than in zone **B**. [1]

....

....

(b) **Image 1.2** represents a section through the leaf of a plant.

Image 1.2



(i) On **Image 1.2**, **draw an arrow** to show the tissue which transports water to all parts of the plant. **Label the arrow with the name of the tissue**. [2]

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(ii) I. Name cells **X** shown in **Image 1.2**. [1]

....

II. State how the stoma and cuticle are involved in the control of water loss from a leaf. [2]

Stoma

....

....

Cuticle

....

....

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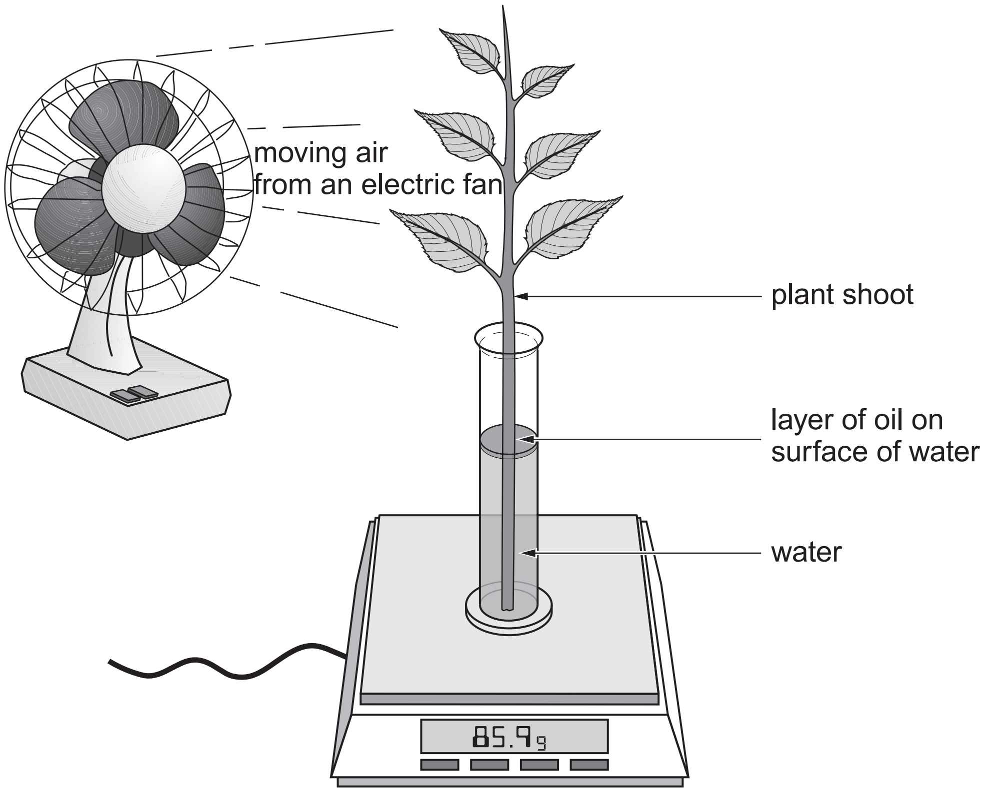
(c) Megan and Rhys investigated the loss of water from a leafy shoot. They used the apparatus shown in **Image 1.3**.

Image 1.3



They recorded the loss of mass after directing moving air, at different speeds, onto the shoot.

(i) State the scientific term for the evaporation of water from the leaves of a plant. [1]

....

(ii) State **one** way in which Megan and Rhys could ensure that they carried out a fair test. [1]

....

....

(iii) State why it was important that the layer of oil was added. [1]

....

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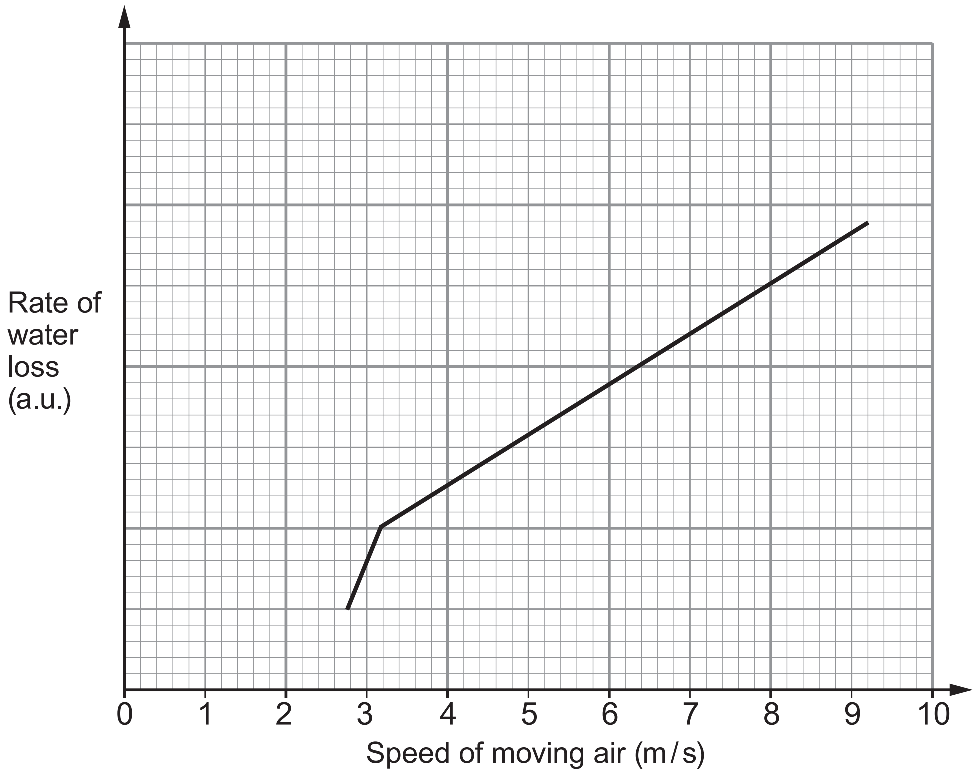
(iv) The results of their investigation are summarised in **Graph 1.4**.

Graph 1.4



I. Describe the effect of increasing the speed of moving air on the rate of water loss. [2]

....

....

....

....

II. **Sketch a line on Graph 1.4** to suggest the result you would expect if the **humidity** of the air was increased. [1]

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